

THRESHOLD SIGNATURE

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Problem Context ?

1. The root of verify signature problem on blockchain
2. Some multi signature scheme to solve the problem
3. Apply such schemes to solve threshold signature problem

Secret Key Knowledge ?

Alice

Private key $x \leftarrow \text{random}$

Public key $P \leftarrow xG$

P

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graph LR; Alice[Alice] -- P --> Bob[Bob]; Alice -- x --> Bob; Bob -- "Ask Alice to prove the ownership of P" --> Bob; Bob -- "Verify xG = P ?" --> Bob; Bob -- "Bob can make proof on Alice's behalf" --> Bob;
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x

Bob

Ask Alice to prove
the ownership of P

Verify $xG = P$?

Bob can make proof
on Alice's behalf

Secret Key Knowledge Proof

Alice

Private key $x \leftarrow \text{random}$

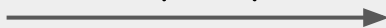
Public key $P \leftarrow xG$

Nonce $r \leftarrow \text{random}$

$R \leftarrow rG$

$s \leftarrow x + r$

(R, s)



Bob

Verify: $sG = P + R ?$

$\Leftrightarrow (x + r)G = P + R ?$

(P, R, s) proves Alice is the owner of private key x

Proof without Private Key

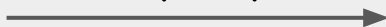
Mallory

Public key P (Alice)

$s \leftarrow \text{random}$

$R \leftarrow sG - P$

(R, s)



Bob

$$sG = P + R ?$$

$$\Leftrightarrow sG = P + sG - P ?$$

$$\Leftrightarrow sG = sG$$

✓ $r \rightarrow R \rightarrow s$

□ $s \rightarrow R$ (without x)

The proof only works when the order of $r \rightarrow R \rightarrow s$ is guaranteed

Interactive Schnorr Signature

Alice

Private key $x \leftarrow \text{random}$

Public key $P \leftarrow xG$

Nonce $r \leftarrow \text{random}$

$R \leftarrow rG$

$s \leftarrow cx + r$

R

c

s

✓ $r \rightarrow R \rightarrow s$

Bob

Challenge $c \leftarrow \text{random}$

Ask Alice ? $sG = cP + R$

Verify $(cx + r)G = cP + R$?

Attack Schnorr Signature ?

Mallory

Public key P (Alice)

Nonce $r \leftarrow \text{random}$

$R \leftarrow rG$

$Q \leftarrow cP + R$

Solve ? $sG = Q$ (DLP)

Bob

Challenge $c \leftarrow \text{random}$

Ask Alice ? $sG = cP + R$

R

c

?

✓ Challenge c will guarantee the order of $r \rightarrow R \rightarrow s$

Non-Interactive Schnorr Signature

Alice

Private key $x \leftarrow \text{random}$

Public key $P \leftarrow xG$

Nonce $r \leftarrow \text{random}$

$R \leftarrow rG$

$s \leftarrow cx + r$

(P, R)

Message (m)

Hash
Function

c

Bob

R

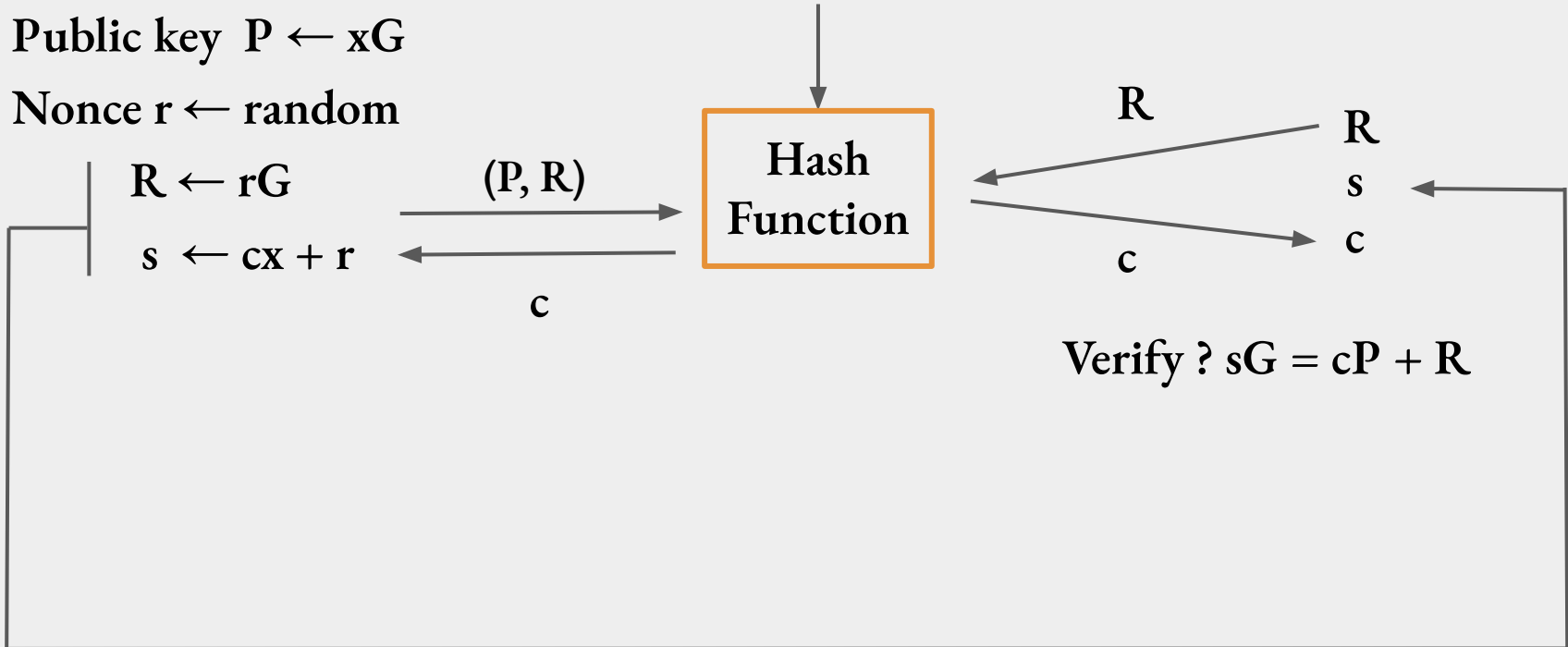
R

s

c

c

Verify ? $sG = cP + R$

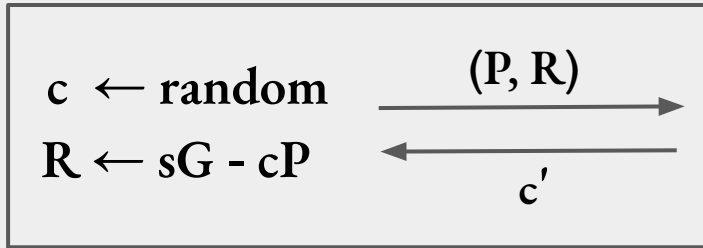


Attack Schnorr Signature ?

Mallory

Public key P (Alice)

$s \leftarrow \text{random}$



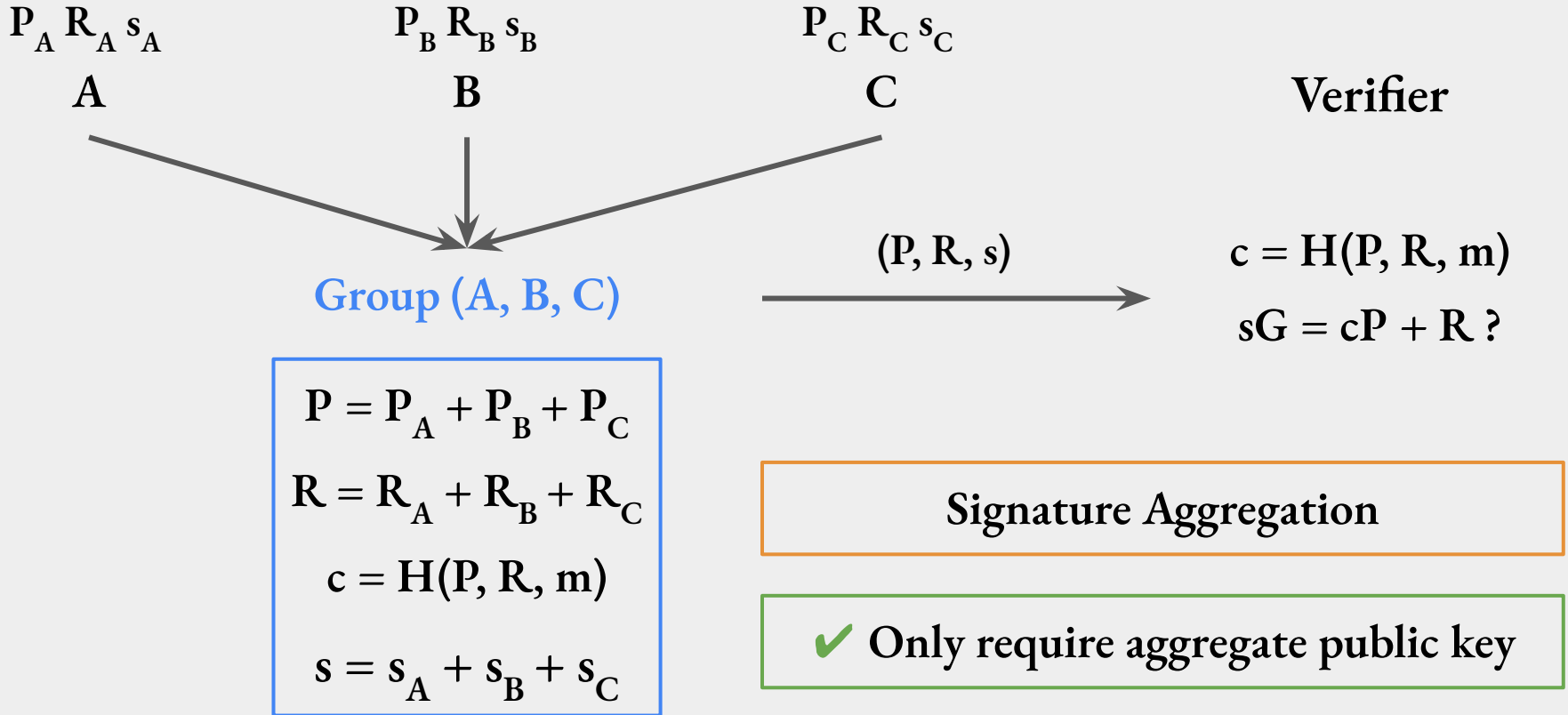
Repeat until ? $c = c'$ (hard)

Message (m)



Bob

Multi Schnorr Signature



Rogue Key Attack

P_A R_A
A

P_B R_B $s_B = cx_B + r_B$
B $P'_B = P_B - P_A$

Verifier

Group (A, B)

(P, R, s)

$$c = H(P, R, m)$$

$$sG = cP + R?$$

$$s_B G = cP_B + R_B$$

$$\begin{aligned} P &= P_A + P'_B \\ P &= P_B \\ R &= R_B \\ c &= H(P, R, m) \\ s &= s_B \end{aligned}$$

□ B can sign on behalf of the group

Create distinct challenge for each member

Bellare - Neven Signature

$$P_A \ R_A \ c_A = H(L, P_A, R, m)$$

$$A \quad s_A = c_A x_A + r_A$$

$$P_B \ R_B \ c_B = H(L, P_B, R, m)$$

$$B \quad s_B = c_B x_B + r_B$$

Group (A, B)

$$\begin{aligned} L &= \{P_A, P_B\} \\ R &= R_A + R_B \\ P &= c_A P_A + c_B P_B \\ s &= s_A + s_B \end{aligned}$$

(L, P, R, s)

Verifier

$$c_A = H(L, P_A, R, m)$$

$$c_B = H(L, P_B, R, m)$$

$$P = c_A P_A + c_B P_B ?$$

$$sG = P + R ?$$

✓ Eliminate Rogue Key Attack

□ Require all public keys to verify challenges

Multi Schnorr

- ✓ Signature Aggregation
- ✓ Aggregated Public Key
- Rogue Key Attack

Bellare Neven

- ✓ Signature Aggregation
- ✓ Eliminate Rogue Key Attack
- Aggregated Public Key

MuSig Signature Scheme

- ✓ Signature Aggregation
- ✓ Aggregated Public Key
- ✓ Eliminate Rogue Key

MuSig Signature Scheme

$$P_1 \ R_1 \ c_1 = H(L, P_1)$$

$$A \quad s_1 = e \cdot c_1 x_1 + r_1$$

$$P_2 \ R_2 \ c_2 = H(L, P_2)$$

$$B \quad s_2 = e \cdot c_2 x_2 + r_2$$

$$P_3 \ R_3 \ c_3 = H(L, P_3)$$

$$C \quad s_3 = e \cdot c_3 x_3 + r_3$$

Group (A, B, C)

$$\begin{aligned} L &= \{P_1, P_2, P_3\} \\ P &= c_1 P_1 + c_2 P_2 + c_3 P_3 \\ R &= R_1 + R_2 + R_3 \\ e &= H(P, R, m) \\ s &= s_1 + s_2 + s_3 \end{aligned}$$

(P, R, s)

Verifier

$$e = H(P, R, m)$$

$$sG = eP + R?$$

$$(s_1 + s_2 + s_3)G = e(c_1 P_1 + c_2 P_2 + c_3 P_3) + R?$$

Secure $r \rightarrow R \rightarrow s$

Prevent Rogue Key

Threshold Signature Scheme

- ✓ Based on MuSig signature scheme
- ✓ Allow minimum of k member to sign on behalf of the group
- ✓ Take advantage of Lagrange Interpolation

$$s = e(\underbrace{c_1x_1 + c_2x_2 + \dots + c_nx_n}_{r_n}) + r_1 + r_2 + \dots +$$

Aggregated Private Key x

Aggregated Public Key P

Lagrange Interpolation

Given a polynomial of order $k - 1$

$$y = f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{k-1}x^{k-1}$$

Polynomial reconstruction:

$$f(x) = \sum_{i=1}^k \left[y_i \prod_{j \neq i}^k (x - x_j)(x_i - x_j)^{-1} \right]$$

Lagrange Interpolation

Example: $y = f(x) = 57 + 23x + 35x^2$

$$y_1 = f(1) = 115$$

$$y_2 = f(2) = 243$$

$$y_3 = f(3) = 441$$

$$\begin{aligned} f(x) &= 115 \left[\frac{x-2}{1-2} \frac{x-3}{1-3} \right] \\ &+ 243 \left[\frac{x-1}{2-1} \frac{x-3}{2-3} \right] \\ &+ 441 \left[\frac{x-1}{3-1} \frac{x-2}{3-2} \right] \end{aligned} \longrightarrow f(0) = 57$$

$$f(0) = \sum_{i=1}^k \left[y_i \prod_{j \neq i}^k (-x_j)(x_i - x_j)^{-1} \right]$$

Lagrange Interpolation

Instead of distribute polynomial roots, each generates a private polynomial and combine secret roots to produce an aggregated secret polynomial

$$f_1(x) = 10 + 6x + 5x^2$$

$$f_2(x) = 13 + 4x + 6x^2$$

$$f_3(x) = 11 + 5x + 7x^2$$

$$f_4(x) = 12 + 3x + 9x^2$$

$$f_5(x) = 11 + 5x + 8x^2$$

$$f_0(x) = 57 + 23x + 35x^2$$

$$f_1(1) \quad f_1(2) \quad f_1(3) \quad f_1(4) \quad f_1(5)$$

$$f_2(1) \quad f_2(2) \quad f_2(3) \quad f_2(4) \quad f_2(5)$$

$$f_3(1) \quad f_3(2) \quad f_3(3) \quad f_3(4) \quad f_3(5)$$

$$f_4(1) \quad f_4(2) \quad f_4(3) \quad f_4(4) \quad f_4(5)$$

$$f_5(1) \quad f_5(2) \quad f_5(3) \quad f_5(4) \quad f_5(5)$$

$$f_0(1) \quad f_0(2) \quad f_0(3) \quad f_0(4) \quad f_0(5)$$

Polynomial Obfuscation

Each member broadcasts the obfuscated polynomial

Everyone confirms the received shared secret roots

$$\boxed{G} \cdot \left[\begin{array}{c|ccc} f_1 & 10 & 6 & 5 \\ f_2 & 13 & 4 & 6 \\ f_3 & 11 & 5 & 7 \\ f_4 & 12 & 3 & 9 \\ f_5 & 11 & 5 & 8 \\ f_0 & 57 & 23 & 35 \end{array} \right] = \boxed{G} \cdot \left[\begin{array}{ccccc} y_{11} & y_{12} & y_{13} & y_{14} & y_{15} \\ y_{21} & y_{22} & y_{23} & y_{24} & y_{25} \\ y_{31} & y_{32} & y_{33} & y_{34} & y_{35} \\ y_{41} & y_{42} & y_{43} & y_{44} & y_{45} \\ y_{51} & y_{52} & y_{53} & y_{54} & y_{55} \\ y_{01} & y_{02} & y_{03} & y_{04} & y_{05} \end{array} \right]$$

Naive Threshold Signature

$$57 \xleftarrow{1} y_{02} \quad y_{04} \quad y_{05} \quad f(0) = \sum_{i=1}^k \left[y_i \prod_{j \neq i}^k (-x_j)(x_i - x_j)^{-1} \right] \quad 2 \quad 3$$

$$\boxed{57} = y_{02} \left[\frac{-4}{2-4} \cdot \frac{-5}{2-5} \right] + y_{04} \left[\frac{-2}{4-2} \cdot \frac{-5}{4-5} \right] + y_{05} \left[\frac{-5}{5-2} \cdot \frac{-4}{5-4} \right]$$

$$s = (\mathbf{x}_1 + \mathbf{x}_2 + \dots + \mathbf{x}_n) + (\mathbf{r}_1 + \mathbf{r}_2 + \dots + \mathbf{r}_n)$$

$$s = (\pi_1 + \pi_2 + \dots + \pi_k) + (r_1 + r_2 + \dots + r_k)$$

Signature Challenges

$x \leftarrow \text{---} y_{02} \quad y_{04} \quad y_{05}$

$$\left[\begin{array}{c|ccc} f_1 & c_1 a_{10} & a_{11} & a_{12} \\ f_2 & c_2 a_{20} & a_{21} & a_{22} \\ f_3 & c_3 a_{30} & a_{31} & a_{32} \\ f_4 & c_4 a_{40} & a_{41} & a_{42} \\ f_5 & c_5 a_{50} & a_{51} & a_{52} \\ f_0 & x & a_{01} & a_{02} \end{array} \right]$$

$$\mathbf{s} = (c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2 + \dots + c_n \mathbf{x}_n) + \Sigma \mathbf{r}_i$$

$$\mathbf{s} = (\pi_1 + \pi_2 + \dots + \pi_k) + \Sigma \mathbf{r}_i$$

MuSig Threshold Signature

Signer

$$s = (e.c_1x_1 + e.c_2x_2 + \dots + e.c_nx_n) + \sum r_i$$

$$s = (e.\pi_1 + e.\pi_2 + \dots + e.\pi_k) + \sum r_i$$

$$P = c_1P_1 + c_2P_2 + \dots + c_nP_n$$

$$R = R_1 + R_2 + \dots + R_k$$

$$e = H(P, R, m)$$

Verifier

$$e = H(P, R, m)$$

$$sG = eP + R ?$$

References

The MuSig Schnorr Signature Scheme

Threshold Signatures and Accountability

Shared Secrets and Threshold Signatures

Key Aggregation for Schnorr Signatures

[Presentation Link on Youtube](#)

[Presentation Slide on Google](#)